Drone Architecture

Jeevan Ram
Manipal, India

***Abstract*—This report studies the major components, sensors, and communication interfaces used in autonomous quadcopters. Their working principles, advantages and limitations are examined, with particular focus on Search and Rescue applications. A proposed system architecture and component-selection analysis are presented to explain the designing of an autonomous drone.**

***Index Terms*—autonomous quadcopter, drone components, sensors, flight control, Search and Rescue**

I. Introduction

Autonomous quadcopters combine propulsion, flight control, sensing, and onboard computation to navigate and perform tasks with limited human intervention. Applications such as Search and Rescue require reliable positioning, obstacle detection, and the ability to identify potential survivors. Achieving these capabilities involves selecting compatible components and understanding their limitations. This report examines the major drone subsystems, compares sensing technologies, and proposes a system architecture and component selection for a Search and Rescue mission.

II. System Architecture

Figure 1 illustrates the proposed system architecture of the autonomous drone for Search and Rescue operations.

III. Electronic Components

A. Flight Controller

It is a embedded computing system which does the calculations to manage a drone's movement and stability.

Main Responsibilities:

- Reading high-frequency inertial and environmental sensor data.
- Filtering noise and estimating the drone's 3D attitude, velocity, and position via an Extended Kalman Filter (EKF).
- Decoding pilot commands form the RC receiver.
- Executing PID (Proportional-Integral-Derivative) control loops and motor mixing algorithms to calculate individual motor speeds.

Stabilization uses a cascaded PID control loop, computing the exact torque corrections needed along the axes:

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau)\,d\tau + K_d \frac{de(t)}{dt} \tag{1}$$

Onboard sensors:

- IMU (accelerometer + Gyroscope): Integrated on the PCB, measure the acceleration on the three axes, measure the rotation on each axes. They should be isolated from the motor vibrations.
- Barometer: Integrated on PCB to measure static air pressure.
- Magnetometer: It's the electronic compass for drone.

Flight modes: Various flight modes are available including acro, stabilise, loiter, altitude hold etc.

Motor control: Motor mixer calculates individual throttle demands for each motor based on the frame geometry from the output of PID loops collective thrust demands.

Failsafe Handling: In case of RC link loss, low battery voltage, depending on the severity and configuration, the FC automatically triggers actions such as Land, Return-to-Launch (RTL) or hold/loiter [1], [2].

B. Electronic Speed Controller (ESC)

Brushes are absent in drone motors. An ESC acts as a DC to three phase AC inverter that regulates speed, direction, and braking of a brushless DC motor (BLDC) based on signals from FC. It uses MOSFETs to switch currect through the motors. Measures the back-emf induced by the spinning magnets and calculates the exact rotor position and fires the next MOSFET.

For the communication between FC and ESC, Pulse Width Modulation (PWM) or Digital signals are used.

Current and Voltage rating:

- Exceeding the voltage rating destroys the MOSFETs and onboard capacitors.
- Continuous current rating (what it can sustain with airflow) and burst current (sustain for a short time).

C. Brushless Motors and Propellers

Brushless DC motors: Drones use outrunner BLDC motors, where the stator sit on the inside and the rotor on the outside spins around it. This allows for more efficiency than Brushed motors and higher torque at lower RPMs but needs precise ESC.

Motor Sizing Nomenclature (XXYY):

- XX – stator diameter in mm.
- YY – stator height in mm.

K_v **Rating (RPM per volt):** Indicates the theoretical unloaded rotational speed of the motor per volt applied:

$$\text{RPM}_{\text{unloaded}} \approx K_v \times V_{\text{battery}} \tag{2}$$

Propeller Geometry:

- Diameter (DD): The total circle diameter traced by the spinning tips. Thrust scales roughly with the swept disc area. Larger diameter propellers are more aerodynamically efficient.

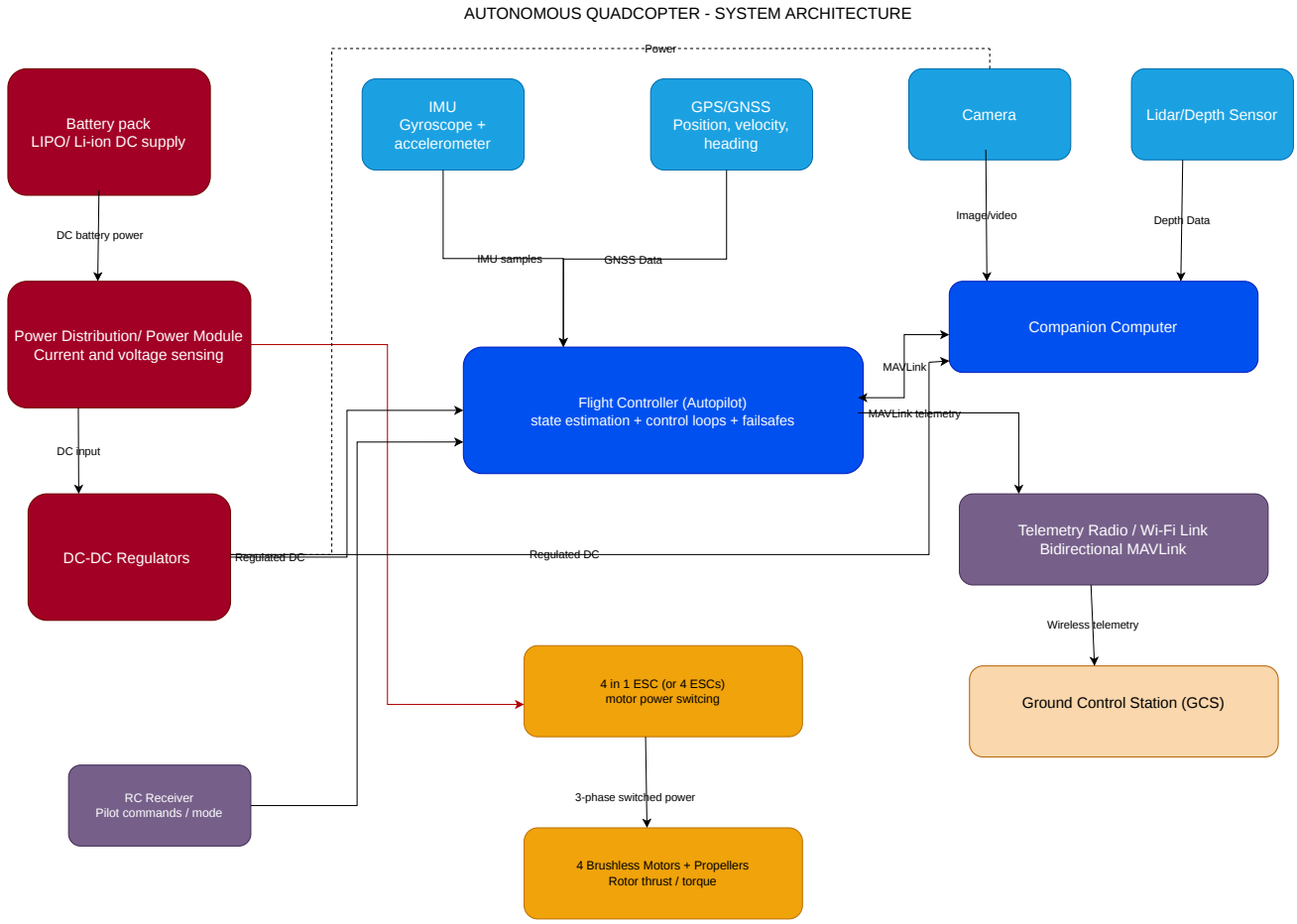


Fig. 1. System architecture of the autonomous Search and Rescue drone.

- Pitch (PP): Theoretical forward distance the propeller would travel in one complete revolution.

Counter-Rotating pairs: Every spinning propeller exerts an reaction torque on the drone frame opposite to it's spin direction. To prevent the drone from spinning uncontrollably in yaw, half the motors spin Clock wise and the other half spin in counter-clockwise using mirrored airfoils so that all propellers push air downward while their reaction torques cancel out to zero during hover [2].

D. Battery and Power System

Battery Chemistry:

- Lithium Polymer (LiPo): Uses a polymer electrolyte that supports high discharge rates (C-Rating), delivering high burst current without severe voltage sag.
- Lithium-ion: Offers higher energy density, but has high internal resistance and lower C-ratings.

Cell Voltage (LiPo):

- Nominal Voltage: 3.7 V/cell
- Fully charged: 4.20 V/cell
- Storage voltage: 3.80–3.85 V/cell
- Safe Discharges Minimum under load: = 3.50 V/cell (dropping below 3.0 V causes irreversible chemical damage and puffing).
- Series and Parallel combinations: Series multiplies voltage. Parallel multiplies capacity and current delivery.
- Capacity (mAh or Ah): Charge storage. Total stored energy in Watt-hours is Wh $=$ Ah $\times$ V_{nominal}. C-Rating: Defines maximum safe continuous current draw:

$$I_{\text{max}} = \text{Capacity (in Ah)} \times \text{C-Rating} \tag{3}$$

[4]

E. GPS and GNSS

GNSS (Global Navigation Satellite System) is the umbrella term for global satellite constellations–including GPS (USA), GLONASS (Russia), Galileo (EU), and BeiDou (China).

The drone's GNSS receiver compares the transmission timestamp with its local arrival time to compute the signal's time of flight (Δt), multiplying by the speed of light (c) to determine the pseudorange ($d = c \cdot \Delta t$) to each satellite. Data form at least 4 satellites allows the receiver to solve for four unknowns via 3D trilateration:

- Latitude (ϕ): Angular distance north or south of the Earth's equator (-90° to $+90^\circ$).
- Longitude (λ): Angular distance east or west of the Prime Meridian (-180° to $+180^\circ$).
- Altitude (h): Height above the WGS84 reference ellipsoid (or converted to Mean Sea Level via a geoid model).
- Receiver Clock Bias (b): Corrects the drift in the drone receiver's inexpensive quartz clock against the satellites' atomic clocks.

RTK (Real-Time Kinematic) GPS uses onboard RTK rover module and stationary Base Station to achieve horizontal accuracy of 0.01 m–0.02 m (1–2 cm) [5].

F. RC Transmitter and Receiver

Provides low latency, direct manual control and emergency override over the UAV.

- RC transmitter transmits the commands over RF.
- RC Receiver mounted on the drone receives the RF signal from the transmitter.

G. Companion Computer

An onboard Linux-based Single Board Computer (SBC) paired with the Flight Controller to handle computationally intensive workloads.

Raspberry Pi 4/5, NVDIA Jetson.

A FC cannot store gigabytes of map data, or execute convolutional neural networks.

IV. Sensors

A. Inertial Measurement Unit (IMU)

- Accelerometer: Measures specific force (linear acceleration plus Earth's gravitational acceleration vector) along three orthogonal axes (X, Y, Z) in m/s^2 or g.
- Gyroscope: Measures angular velocity (rate of rotation) around the three body axes (Roll rate p, Pitch rate q, Yaw rate r) in deg/s or rad/s.

Gyroscope Flaw (Integration Drift): To find the drone's tilt angle from a gyroscope, the FC integrates angular velocity over time ($\theta(t) = \theta_0 + \int \omega \, dt$). Even tiny sensor biases or thermal noise accumulate continuously during integration, causing the estimated angle to drift several degrees per minute.

Accelerometer Flaw (Vibration Noise): An accelerometer can measure absolute pitch and roll tilt relative to Earth's $1g$ gravity vector ($\theta = \arctan(a_x/\sqrt{a_y^2 + a_z^2})$) without drifting over time. However, spinning propellers and linear accelerations corrupt the gravity reading with high-frequency vibration noise, making raw accelerometer angles unusable over short timescales.

An Extended Kalman Filter (EKF) or Complementary Filter applies a high-pass filter to the integrated gyroscope (trusting it for fast, short-term changes) and a low-pass filter to the accelerometer (trusting it to anchor long-term gravity reference and eliminate gyro drift):

$$\theta_{\text{est}} = \alpha(\theta_{\text{est, prev}} + \omega_{\text{gyro}} \cdot \Delta t) + (1 - \alpha)\theta_{\text{accel}} \tag{4}$$

[6]

B. Magnetometer

A 3-axis magnetometer measures the strength and direction of the ambient magnetic field vector (in Gauss or microteslas, μT) along the drone's X, Y, Z body axes.

C. Barometer

A MEMS pressure sensor containing a microscopic silicon diaphragm that flexes in response to changes in ambient static atmospheric pressure (measured in Pascals or hectopascals, hPa).

D. LiDAR

Emits rapid pulses of near-infrared laser light and uses a photodetector to measure the exact Time-of-Flight for the reflected photon pulse to return from a surface:

$$d = \frac{c \cdot \Delta t}{2} \tag{5}$$

where $c \approx 3 \times 10^8$ m/s is the speed of light and the factor of 2 accounts for the round-trip path.

Applications: Directly measures precise distances to objects like trees, walls and wired for real time collision avoidance.

E. Depth Camera

A standard 2D camera captures only the colour intensity per pixel, a depth camera produces a Depth Map (Z-buffer) where every pixel encodes the perpendicular distance from the camera plane to the surface using two horizontally separated cameras and triangulation.

F. Optical Flow Sensor

It points a small downward facing, high frame rate monochrome camera at the ground and tracks the apparent velocity of brightness patterns moving across successive image frames.

The sensor tracks pixel patches between frames and uses algorithms such as Lucas-Kanade optical flow or block matching to measure angular pixel velocity.

V. Additional Sensor Research for Autonomous Drones

A. Millimeter-Wave (mmWave) Radar

Millimeter-wave radar is an active sensing technology that transmits radio waves and analyses the signals reflected by surrounding objects. Frequency-modulated continuous-wave (FMCW) radar transmits chirps whose frequency varies with time. The frequency difference between the transmitted and received signals is used to estimate target range, while Doppler processing estimates radial velocity. Antenna-array measurements can also provide the direction of detected objects. For example, Texas Instruments' IWR6843 operates in the 60–64 GHz band, whereas the AWR1843 operates in the 76–81 GHz band; these are different devices, not one continuous operating range [7], [9], [10].

On an autonomous drone, radar measurements can support obstacle detection, relative-motion estimation, and navigation

in environments where visible-light cameras are impaired by darkness, smoke, or haze. Direct radial-velocity measurements can help distinguish stationary surroundings from approaching or moving objects. Compared with cameras and LiDAR, radar can be more robust in some poor-visibility conditions and can provide useful motion information without relying solely on tracking objects across image frames [7].

B. Long-Wave Infrared (LWIR) Thermal Camera

A thermal camera detects infrared radiation emitted by objects rather than depending on visible light reflected from them. Long-wave infrared cameras commonly operate in the 8–14 μm wavelength band. In an uncooled microbolometer detector, absorbed infrared radiation changes the temperature and electrical resistance of microscopic sensing elements. The electronics process these changes to produce a thermal image. Some camera models also support calibrated temperature measurements, but this capability depends on the specific device and configuration [8].

For search-and-rescue drones, thermal imaging can help identify people or other warm targets at night or in low-visible-light environments. It can also assist in locating fire hotspots and detecting overheating equipment. Unlike a conventional RGB camera, a thermal camera does not require visible illumination, making it useful when darkness limits ordinary imaging.

VI. Component Selection for Search and Rescue Drone

The components are selected to balance reliability, computational capability, weight, and cost for an autonomous search-and-rescue (SAR) quadcopter.

A. Flight Controller

Selected component: Holybro Pixhawk 6C running PX4.

The Pixhawk 6C provides real-time attitude stabilization, sensor integration, waypoint navigation, and failsafe handling. Its redundant IMUs and dedicated interfaces for GPS, RC, and telemetry support reliable outdoor operation [11].

B. Companion Computer

Selected component: NVIDIA Jetson Orin Nano.

The Jetson Orin Nano provides GPU-accelerated processing for onboard computer vision and deep-learning models. It can process camera images to detect people and support autonomous navigation by exchanging commands and vehicle-state information with the flight controller through MAVLink. Its power consumption and payload weight must be considered during airframe selection [12].

C. Positioning System

Selected technology: Multi-constellation GNSS receiver with an external compass.

A GNSS receiver provides outdoor position and velocity estimates for waypoint navigation, while the compass supplies heading information. A compatible module such as the u-blox NEO-M9N can receive signals from multiple satellite constellations. GNSS accuracy can degrade near buildings, under dense tree cover, or in obstructed environments; therefore, it should be supplemented with inertial sensing and, where necessary, visual or other local positioning [13].

D. Obstacle-Detection Sensor

Selected component: Benewake TFmini-S LiDAR.

The TFmini-S measures distance using time-of-flight ranging and offers a compact, lightweight solution for detecting nearby obstacles. Its specified measurement range reaches 12 m for highly reflective targets, with shorter range for less reflective surfaces. However, it is a single-point sensor, not a scanning LiDAR, so multiple sensors or a depth camera may be required to detect obstacles across a wider field of view [14].

E. Human-Detection Sensor/Camera

Selected component: Raspberry Pi Camera Module 3, connected to the companion computer.

The camera captures high-resolution images that can be processed by a trained person-detection model on the Jetson Orin Nano. This approach enables onboard human detection without continuously transmitting raw video to the ground station. Detection performance depends on lighting, distance, occlusion, camera orientation, and model accuracy; thermal imaging may be added for night-time or low-visibility searches [15].

F. Communication System

Selected technology: MAVLink-compatible telemetry radio link between the flight controller and Ground Control Station.

A telemetry radio enables the ground station to receive position, battery status, flight mode, and other vehicle telemetry, while supporting mission uploads and parameter configuration. MAVLink provides the message protocol, while the radio provides the wireless transport. A separate Wi-Fi or cellular link may be used to transmit higher-bandwidth images when coverage permits. Communication range and reliability depend on antenna placement, radio regulations, interference, and terrain.

References

[1] ArduPilot documentation. Available: `https://ardupilot.org/`

[2] Q. Quan, *Introduction to Multicopter Design and Control*. Springer, 2017. Chapters 4, 6, and 10 cover flight controller hardware architecture, attitude representation, and cascaded PID control.

[3] Q. Quan, *Introduction to Multicopter Design and Control*. Springer, 2017.

[4] D. Linden and T. B. Reddy, Eds., *Linden's Handbook of Batteries*, 4th ed. McGraw-Hill.

[5] E. D. Kaplan and C. J. Hegarty, *Understanding GPS/GNSS: Principles and Applications*, 3rd ed. Artech House, 2017.

[6] W. T. Higgins, "A comparison of complementary and Kalman filtering," *IEEE Transactions on Aerospace and Electronic Systems*, 1975.

[7] C. Iovescu and S. Rao, "The fundamentals of millimeter wave radar sensors," Texas Instruments, White Paper SPYY005A, 2020.

[8] M. Vollmer and K.-P. Möllmann, *Infrared Thermal Imaging: Fundamentals, Research and Applications*, 2nd ed. Wiley-VCH, 2017.

[9] Texas Instruments, "IWR6843 product documentation."

[10] Texas Instruments, "AWR1843 product documentation."

[11] PX4 Autopilot, “Holybro Pixhawk 6C,” *PX4 User Guide*. [Online]. Available: https://docs.px4.io/main/en/flight_controller/pixhawk6c
[12] NVIDIA, “Jetson Orin Nano Developer Kit User Guide.” [Online]. Available: https://docs.nvidia.com/jetson/orin-nano-devkit/user-guide/
[13] u-blox, “NEO-M9N GNSS Module,” product documentation. [Online]. Available: https://www.u-blox.com/
[14] Benewake, “TFmini-S LiDAR Datasheet.” [Online]. Available: https://ai.benewake.com/en/resources/technical-documents/tfmini-s-datasheet/
[15] Raspberry Pi Ltd., “Camera Module 3 Product Brief.” [Online]. Available: https://datasheets.raspberrypi.com/camera/camera-module-3-product-brief.pdf